INDEX

NAME: Sneha Raju STD.: ____ SEC.: ____ ROLL NO.: 519 SUB.: Computer networks observation

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Date: 20/7/26

STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

AIM: To study the various network commands used in linux and windows.

1) arp -a : ARP stands for address resolution protocol. It shows the IP address and MAC address of your router. (like address book)
(fedora)

output: -gateway (10.0.2.2) at 52:55:0a:00:02:02
MAC address
[ether] on enp0s3
private IP used within an organisation (IP address)

2) hostname : It is the simplest TCP/IP command. It displays the name of your computer. (fedora)

output: fedora

3) ipconfig/all : Displays configuration information about router, gateway, DNS, DHCP. (On windows cmd, not fedora, does not work on fedora)

output: Windows IP Configuration

Hostname . . . . : DESKTOP-2NCTKD3

Primary Dns Suffix :

Node Type : Mixed

IP Routing enabled : NO

WINS Proxy enabled : NO

4)

Physical address . . . . : F8-ED-FC-5C-07-F9

DHCP v6 IAID . . . . : 100199932

IPV4 address . . . . : ~~192.168.561~~ 172.16.10.139 (Preferred)

Subnet mask . . . . : ~~255.255.252.0~~

Physical
DHCP
IPV4
Subit

DNS → www.google.com
converts to IP
and gives to server

4)

nbtstat -a : Helps solve problems with NetBios name resolution.
(cmd-windows)

output: Displays protocol statistics and current TCP/IP connections using NBT

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

-a (adapter status) Lists the remote machine's name table given its name

.
.
.
.

5) (fedora)
net stat : Displays variety of statistics about computer's active TCP/IP connections.

Output: Active Internet Connections (w/o servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign address	state
~~tcp~~ udp ~~tcp~~	3 0 3	0	fedora: bootpc	- gateway: bootps	ESTAB-LISHED

Active UNIX domain sockets (w|o servers)

:
:
:

6) netstat -r

output: kernel IP routing table

Destination	gateway	Gen Mask	Flags	MSS Window	irtt	Iface
default	-gateway	0.0.0.0	UG	0 0	0	enp0s3
10.0.2.0	0.0.0.0	255.255.255.0	U	0 0	0	enp0s3

6) nslookup : Tool used to perform DNS lookups in Linux.
(fedora)

nslookup www.google.com

output: Server: unknown

Address: 172.16.8.1

Non authoritative answer:

Name: www.google.com

Addresses: 2001:4860:4827:7700::

7) pathping (cmd) : It traces the route to the destination address then launches a 25 second test of each router along the way.

output: Usage: pathping [-g host-list] [-h maximum-hops]
[-i address] [-n] [-p period] [-q num-queries]
[-w timeout] [-4] [-6] target-name

Options:

-g host-list : Loose source route along host-list

8) ping : It is used to test the connectivity between (fedora) 2 nodes. It uses ICMP to communicate to other devices.

output: ping [-t] [-a] [-n count] [-l size] [-f]
[-i TTL] [-v TOS] [-r count] [-s count]
[[-j host-list] | [-k host-list]] [-w timeout]
[-R] [-S Srcaddr] [-c compartment] [-p]
[-4] [-6] [target-name]

:

eg. ping localhost

output: Pinging DESKTOP-2NCTK03 [::1] with 32 bytes of data

Reply from ::1: time < 1ms

Reply from ::1: time < 1ms

:

9) route : It is used to show / manipulate the IP routing (fedora) table. It is used to setup static routes to specific host or networks via an interface.

output: Manipulates network routing tables

ROUTE [-f] [-p] [-4]-6] command [destination]

[MASK netmask] [gateway] [METRIC metric]

[IF interface]

kernel IP Routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
default	-gateway	0.0.0.0	UG	100	0	0	enp0s3
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0	0	enp0s3

IMPORTANT LINUX NETWORKING COMMANDS:

1) ip: It can show address information, manipulate routing plus display network various devices, interfaces and tunnels.

ip <OPTIONS> <OBJECT> <COMMAND>

a) To show the IP address assigned to an interface on your server

ip address show

o/p: lo: <LOOPBACK, UP, LOWER-UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

link/loopback 00:00:00:00 brd 00:00:00:00

inet 127.0.0.1/8 scope host lo

b) To assign IP to an interface:

ip address add 192.168.1.254/24 dev enp0s3

O/P:

enp0s3 : <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000

inet 192.168.1.254/24 scope global enp0s3
valid_lft forever preferred_lft forever

how to execute?

sudo ip address show
[sudo] password for student : rec
sudo -i
changes to root.
ip address add 192.168.1.254/24 dev enp0s3
ip address show

c) To delete IP on any interface:

→ ip address del 192.168.1.254/24 dev enp0s3
→ ip address show

O/P: enp0s3 : <BROADCAST, MULTICAST, UP, LOWER-UP> mtu 1500 qdisc noqueue state UNKNOWN group default qlen 1000

inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3

valid_lft 85125 sec preferred_lft 85125sec

2) ifconfig - It is used for configuring and troubleshooting networks.

O/P : enp2s0 : flags = 4163 <UP, BROADCAST, RUNNING, MULTICAST> mtu 1500

inet 127.16.9.200 netmask 255.255.252.0 broadcast 172.16.11.255

:

3) mtr : It serves as a network diagnostic and troubleshooting tool. It combines the functionality of ping and traceroute commands.

mtr <options> hostname/IP

→ mtr google.com

O/P : My traceroute [v0.95]

fedora (172.16.9.200) → google.com (142.251.222.206)

~~Pang Bit Pattern~~ :

Keys: Help Display mode Restart statistics Order of fields qu

Host 2026-07-21T06:35

		Packets				Pings			
1.	(waiting for reply)								
2.	137.97.163.129	Loss%	Snt	Last	Avg	Best	Wrst	StDev	
3.	(waiting for reply)	0.0%	109	1.7	3.1	1.2	18.9	3.2	

→ mtr -b google.com

O/P: My traceroute [v0.95]

fedora (172.16.9.200) → google.com (142.250.66.14)

keys: Help Display mode Restart statistics Order of fields quit

4) tcpdump : It is used for capturing and displaying packets.

→ tcpdump -D - Interfaces tcpdump

Output:

1. enp0s3 [up, running connected]
2. any [pseudo-device that capture on all interface) [up, Running]
3. lo [up, running, Loopback]...

→ dnf install -y tcpdump - command to install tcpdump

→ tcpdump -i enp0s3 - Capture traffic on enp0s3

Output:

dropped privs to tcpdump

tcpdump: verbose output supressed...

09:12:04:650915 ip6 gateway > ff02::1:ICMP6

route advertisement, length 56...

→ tcpdump -i enp0s3 -c 10 : Capture traffic to and from the host

→ Output:

09:17.36.131596 IP6 fedora.mdns > ff05:fb mDNS:

0 PTR (QM)? _raop._tcp.local (34)

10 packets captured

10 packets received by filter

0 packets dropped by kernel

5) ping: It verifies IP-level connectivity to another TCP/IP computer by sending ICMP echo request messages.

→ ping google.com

O/P: PING google.com (142.251.221.174) 56 (84) bytes

64 bytes from bhmaa-au-in-f14.1e100.net

(142.251.221.174): icmp_seq=1 ttl=119 time=3.69 ms

CONFIGURING AN ETHERNET CONNECTION USING NMCLI

1. nmcli connection show

O/P:	NAME	UUID	TYPE	DEVICE
	Wired connection 1	68116511 8-eba5	ethernet	enp2s0
	TLFR5-193	19519 4d8-d694	wifi	--

IP command continuation

d) Alter the status of the interface by bringing the interface enp0s3:

→ ip link set enp0s3 up

→ ip link show enp0s3

O/P: 2: enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500 qdisc fq_codel state UP mode Default group default qlen 1000

link/ether 08:00:27:9e:65:ad brd ff:ff:ff:ff:ff:ff

altname enx080027eb5ad

c) After the status of the interface by bringing down interface enp0s3:

→ ip link set enp0s3 down

→ ip link show enp0s3

O/P: enp0s3 : <BROADCAST, MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode group default qlen 1000

link/ether 08:00:27:9e:65:ad brd ff:ff:ff:ff:ff:ff

altname enx080027eb5ad

b) After the state of the interface by enabling promiscuous mode for enp0s3:

→ ip link set enp0s3 promisc on

→ ip link show enp0s3

O/P: enp0s3 : <BROADCAST, MULTICAST, PROMISC> mtu 1500 qdisc fq_codel state DOWN mode default group default qlen 1000

link/ether 08:00:27:9e:65:ad brd ff:ff:ff:ff:ff:ff

g)

PING COMMAND CONTINUED

ping -c 10 google.com

PING google.com (142.250.77.110) 56 (84) bytes of data

o/p: 64 bytes from lcmaaa-al-in-f14.1e100.net

(142.251.223.238) : icmp-seq=1 ttl=255 time = 6.85 ms

Configuring an ethernet connection by using nmcli:

Procedure

1. List the network manager connection profiles

nmcli connection show

o/p: NAME UUID TYPE DEVICE

Wired connection 1 06726542-1b12-... ethernet enp0s3

2. nmcli connection add con-name <connection-name>

ifname <device-name>

modify an existing profile

3. Rename the connection profile

nmcli connection modify wired connection

4. Display current settings of connection profile

→ nmcli connection show

NAME	UUID	TYPE - DEVICE
enp0s3	a1bfd672-888a... 33b4-bb5b-5f6089...	
lo	eady ...	

nmcli connection up internal-LAN

O/P: Connection successfully activated CD-Bus active path : /org/freedesktop/NetworkManager/

Active connection/3

3/8/26

Result: Thus various commands used in linux and windows are executed successfully.

3/8/26

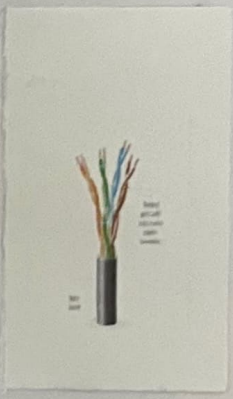
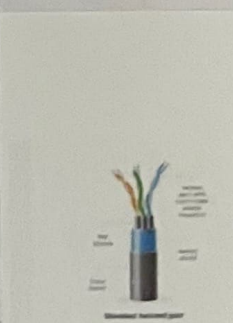
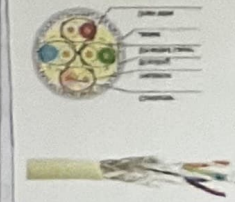
PRACTICAL - 2.

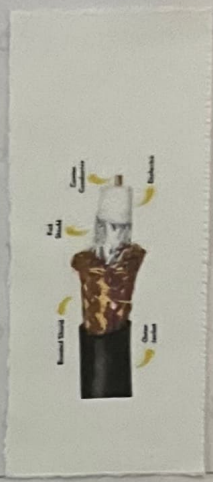
Aim : Study of different types of Network cables

a) understand different types of network cables.

Different type of cables used in networking are :

1. Unshielded Twisted Pair (UTP) Cable.
2. Shielded Twisted Pair (STP) Cable.
3. Coaxial cable
4. Fibre Optic cable.

CABLE TYPE	CATEGORY	MAX. DATA TRANSMISSION	ADVANTAGES / DISADVANTAGES	APPLICATION / USE	IMAGE
UTP	Category 3	10bps	ADVANTAGES • Cheaper in cost • Easy to install as they have a smaller overall diameter. DISADVANTAGES • More prone to (EMI) Electromagnetic interface & noise.	10 Base - T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet Gigabit Ethernet	
	Category 5	Upto 100 Mbps			
	Category 5e	1 Gbps			
STP	Category 6, 6a	10Gbps	ADVANTAGES • Shielded • Faster than UTP • Less susceptible to noise and interference DISADVANTAGES • Expensive • Greater Installation Effort	Gigabit Ethernet 10G Ethernet (55m) widely used in data centres Gigabit Ethernet 10G Ethernet (100m)	 
SSTP	Category 7	10Gbps			

Cable Type	Category	Max. Data Transmission	Advt/ Disadvantages	Application/ Use	Image
UTP	Category 3	10bps	Advantages: • cheaper cost • Easy to install as they have a small Disadvantages Hd distance cost Size is bulky	Speed of Signal is 500 m TV network High speed internet connections.	
fibre optics Cable	Single mode Multi mode	100Gbps	Advantages • High speed • High band width • High Security • Long distance Disadvantages • Expensive • Skilled installery	• Max distance of fibre optics cable is around 100 metres.	

• Make your own Ethernet cross-over cable / Straight cable.

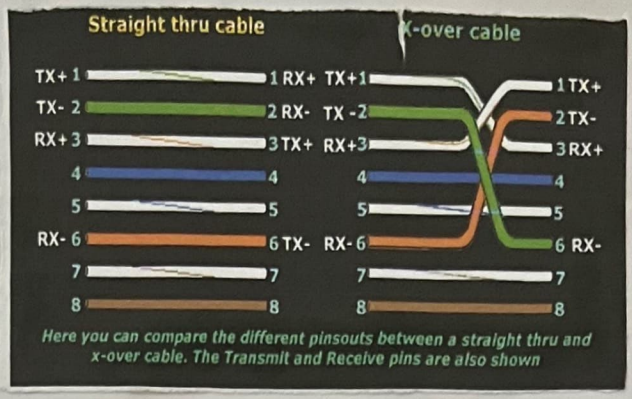
Tools & parts needed:

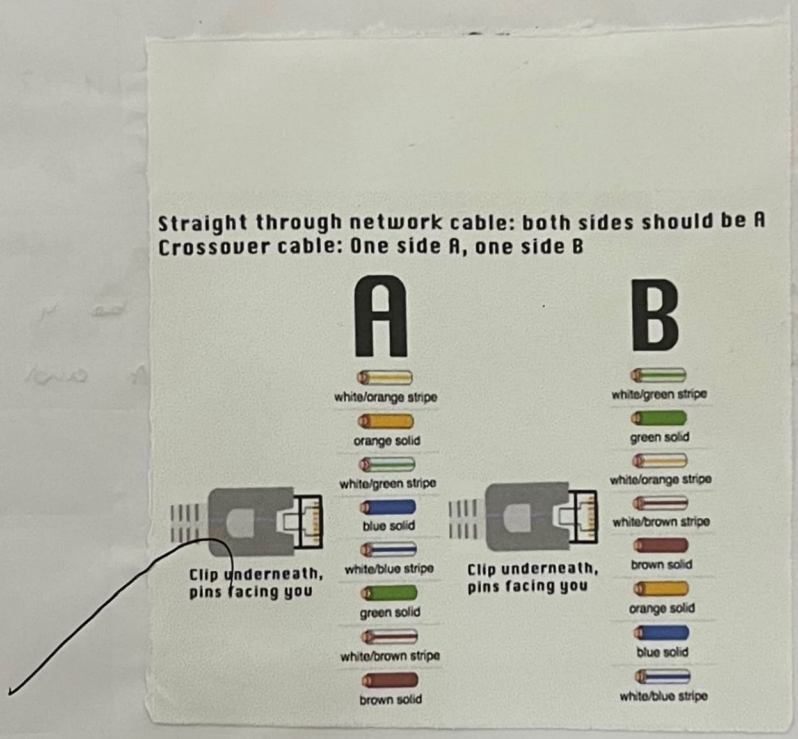
• Ethernet cabling. CAT 5E is certified for gigabit support, but CAT 5 cabling works as well, just over shorter distances.

• A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the cables

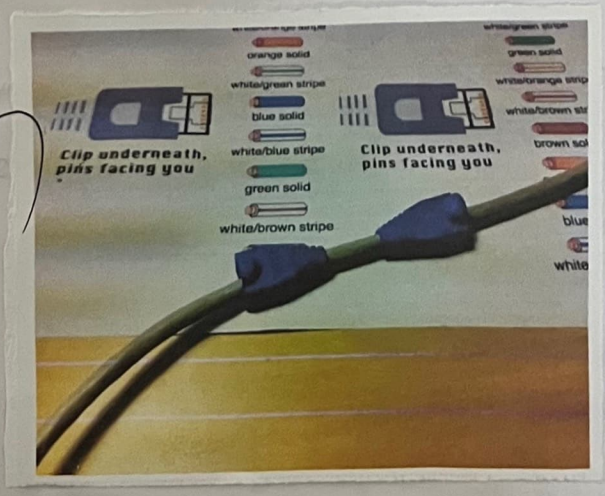
• Two RJ45 plus

• Optional & plug shields

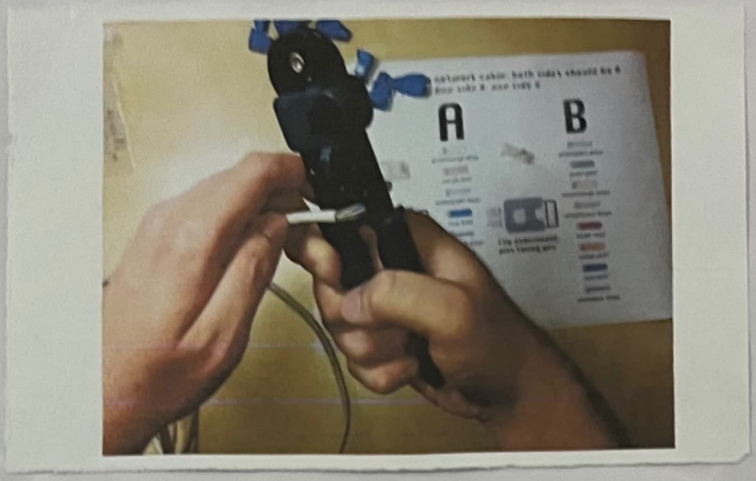




Step1 : To start construction of the device, begin by threading shields onto the cable.



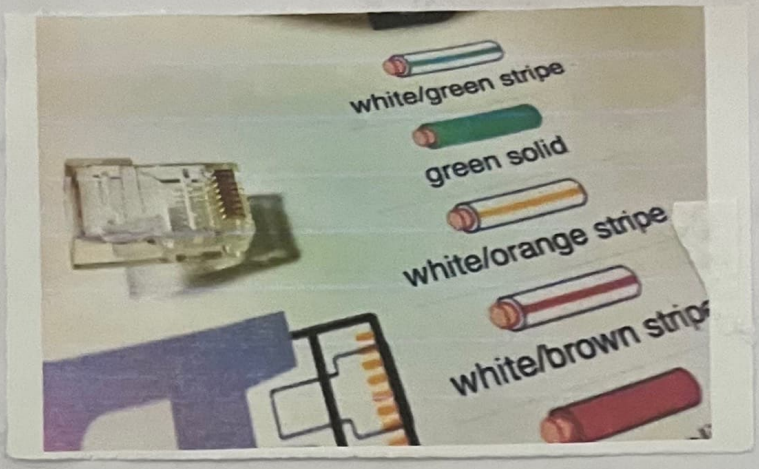
Step 2: Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area.



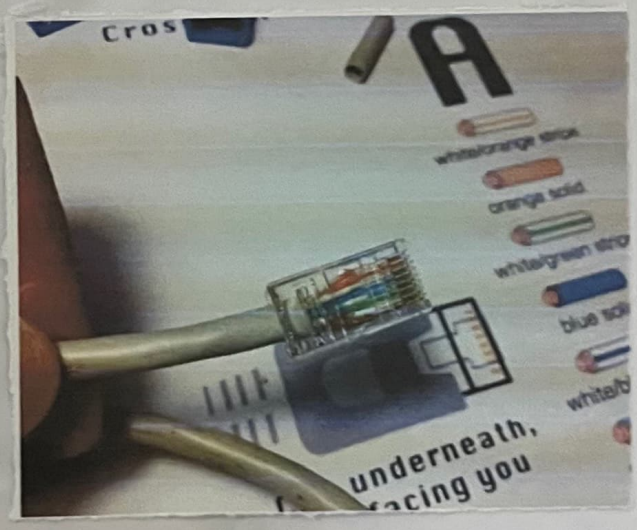
Step 3: Untangle the wires. There should be 4 twisted pairs. One must be in arrangement A and another in B.



Step 4: Once the order is correct, bunch them together in a line. If any stick out, snip them back to create an even level.



Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it hasn't, it means too much shielding has been stripped off.



Step 6: Insert into the crimping tool & push down.

Step 7: Repeat for the other end using diagrams.

To test, plug it in and connect 2 devices directly.

Result: Thus, the different types of network cables were studied.

Date: 4/8/26

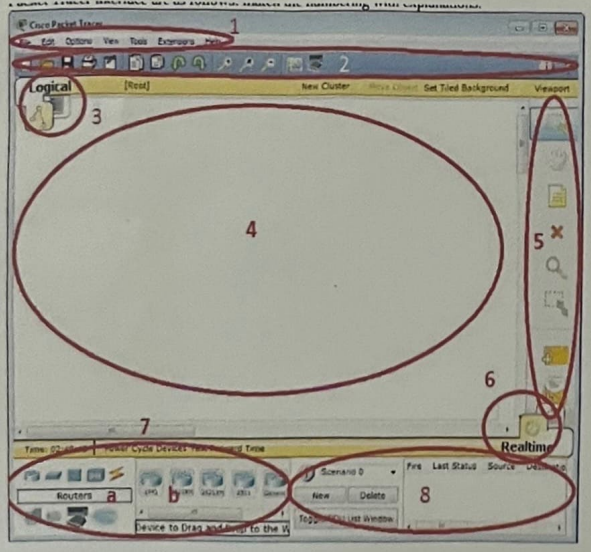
Practical 3

Aim: To Study packet tracer tool installation & user interface overview.

Introduction :

- packet tracer allows you to model, complex systems without dedicated equipment
- It helps you practice your network configuration & trouble shooting

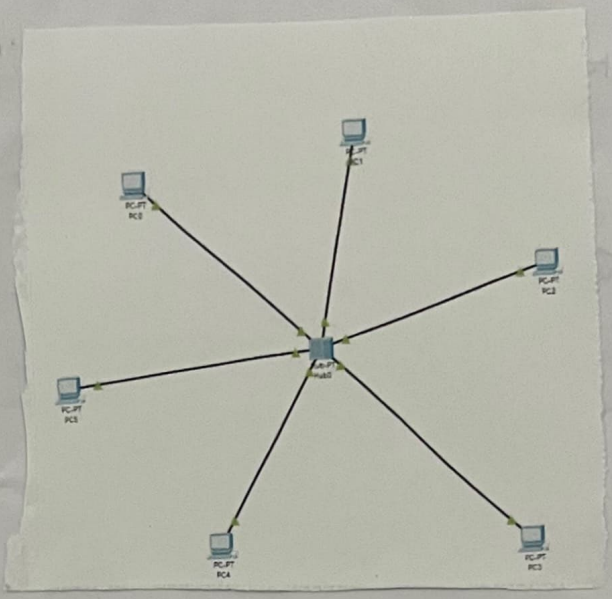
User Interface Overview:



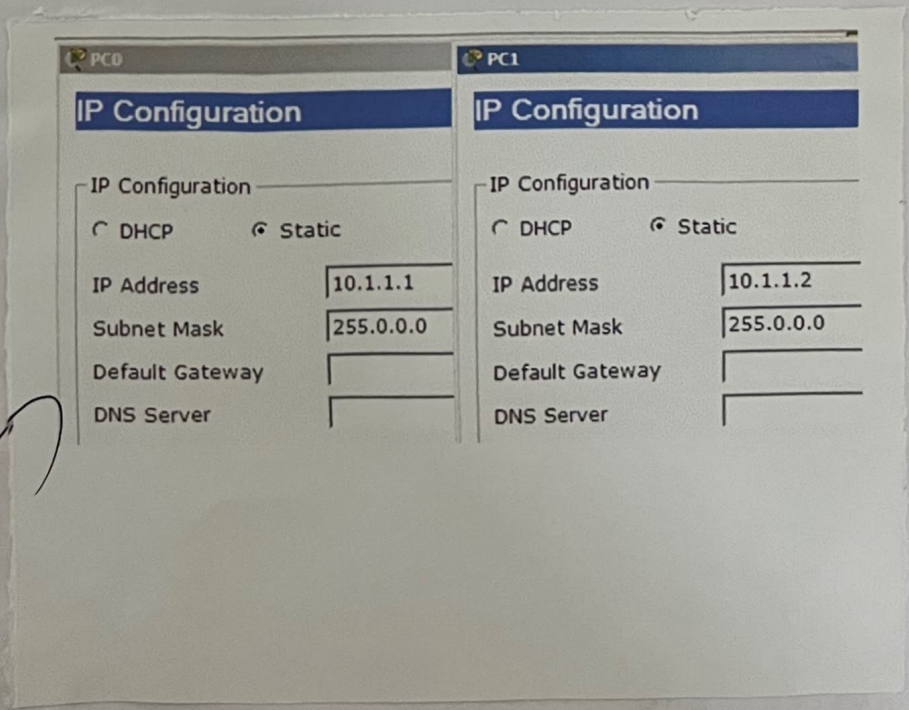
Analyse the behaviour of network devices using CISCO Packet Tracer simulator.

1. A. H Generic PCs & 1 HUB
   B. H Generic PCs & 1 Switch

2. Click on copper Straight through cable & connect all the 4 PCs to the Hub & the Switch.



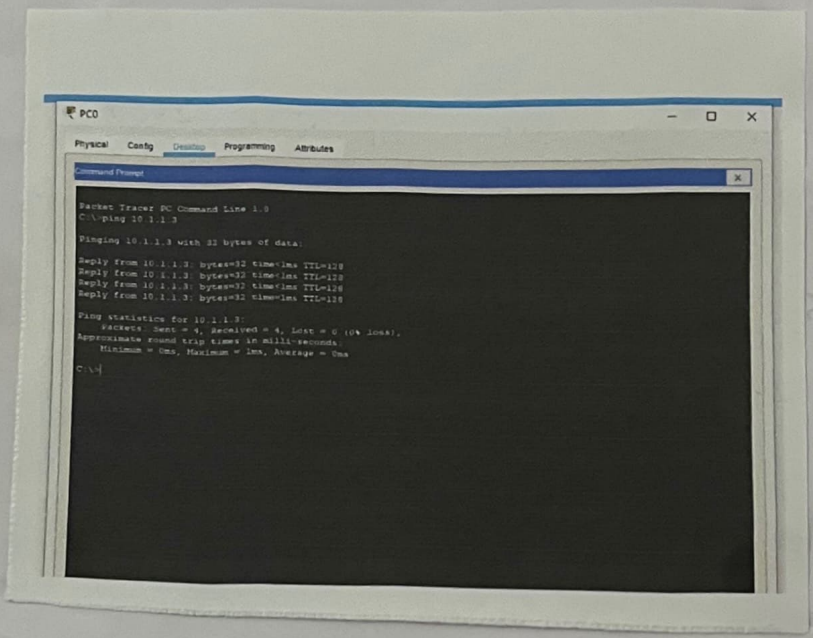
3. Click on the PCs connected to the HUB, Go to the Desk Top TAB & Click on IP configuration. Enter an IP addres & SUBNET MASK.

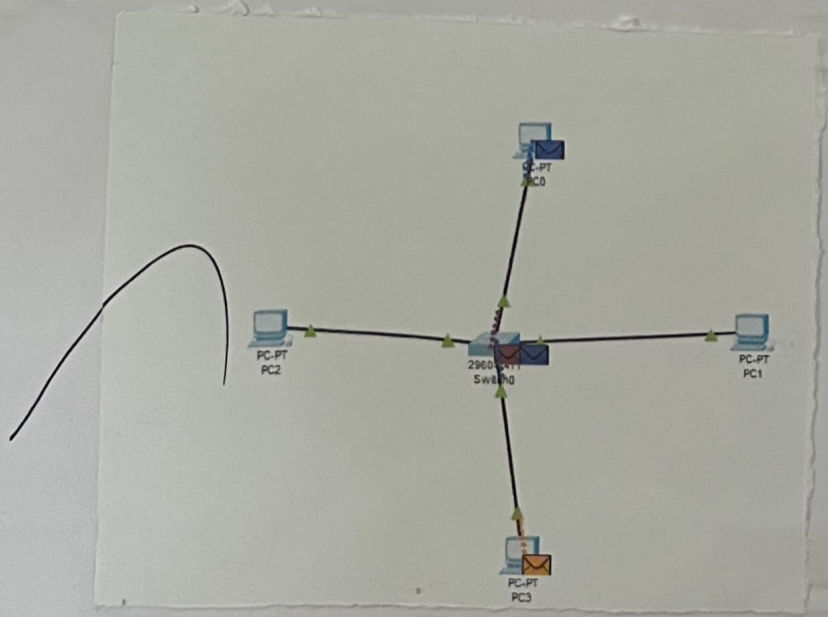


4. Click on the PDU & observe its flow from the source Pc to destination Pc

5 Repeat step 3 to step 5 for the other PCs connected to the switch

6 observe how HUB & switch are forwarding the PDU.





Result: Thus, the CISCO packet tracer has been installed and studied.

10/8/26

Practical - 4.

AIM : Setup and Configure a LAN (Local area network) using a Switch and Ethernet Cables in your lab.

How to setup a LAN

Step. 1. Plan and Design an appropriate network topology taking into account network requirements and equipment location.

Step- 2. you can take 4 Computers, a switch, with 8, 16, or 24 ports which is sufficient for networks of these sizes, and 4 Ethernet Cables.

Step. 3. Connect your Computers to network Switch via an Ethernet Cable, which is as Simple as plugging one end of the ethernet Cable into your Computer and the other end into your network switch.

Step- 4. Assign IP address to your PCs.

1. Log on to the Client Computer as Administrator or as Owner.
2. Click Network and Internet Connections.
3. Right click Local Area Connection / Ethernet -> Go to Properties -> Select Internet Protocol (TCP/IP V4) -> Click on properties -> Select use the following ip address option and assign ipaddress.

Step- 5. Configure a network switch:

1. Connect your Computer to the switch: To access the switch's web interface, you will need to connect your Computer to the switch using an Ethernet Cable.

2. Log into the web interface: Open a web browser and enter the IP address of the switch in the address bar. This should bring up the login page for the switch's web interface. Enter the username and password to log in.

3. Configure basic settings: Once you are logged in, you will be able to configure basic settings for the switch.

4. Assign IP address as: 10.1.1.5, Subnet mask 255.0.0.0.

Step 6: Check the connectivity between switch and other machine by using ping Command in the Command prompt of the device.

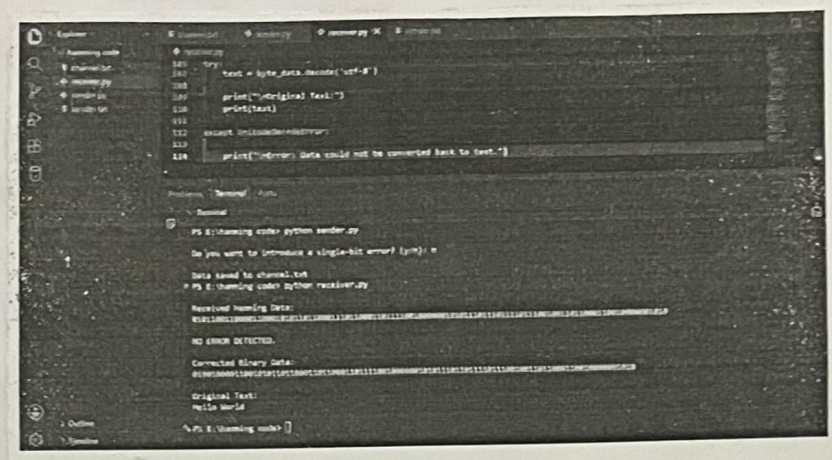
Step 7. Select a folder, -> go to properties -> Click Sharing tab -> Share it with everyone on the Same LAN.

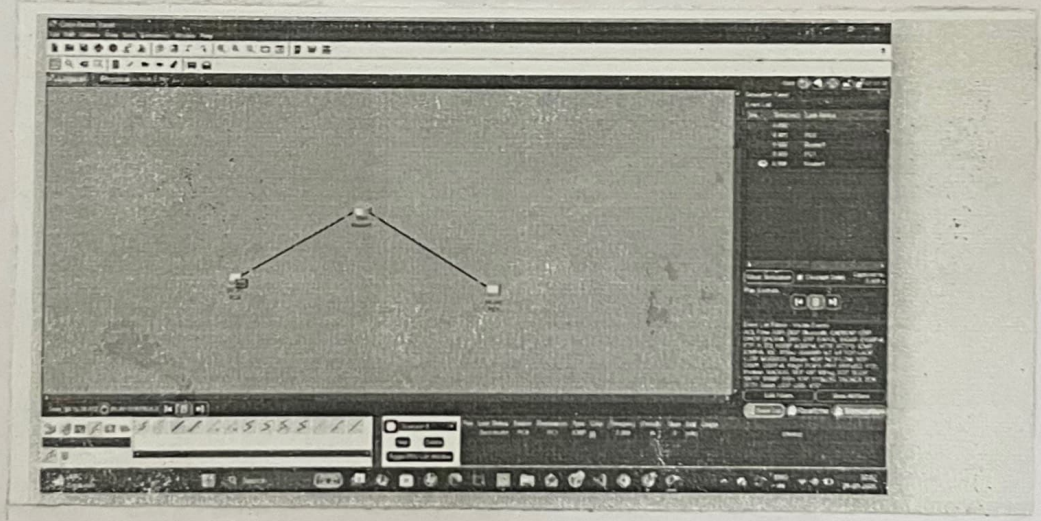
Step 8: Try to access the Shared folder from others Computers of the network.

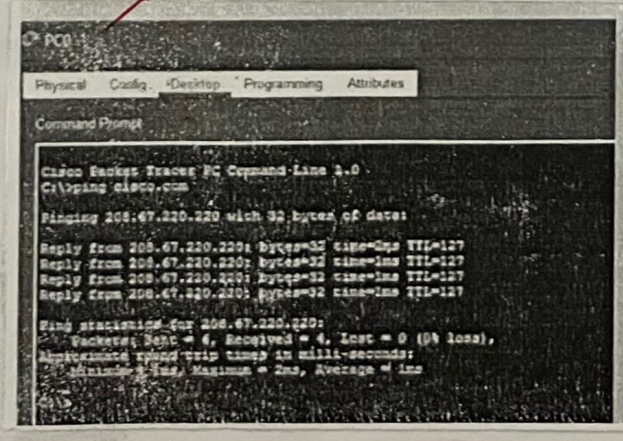
PRACTICAL - 5.

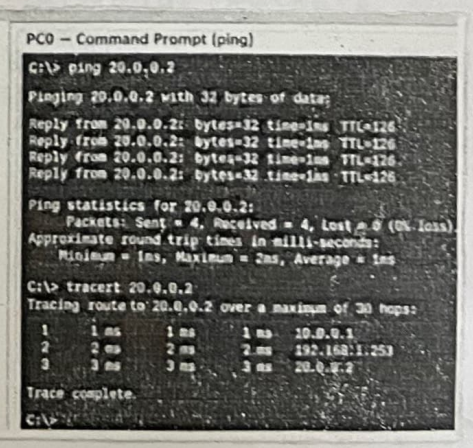
AIM : Experiments on Packet capture tool : Wireshark.

Packet Snif









Date: 18/8/26 Practical - 5

AIM Experiments on Packet capture tool : Wireshark.

Packet Sniffer

- Sniff messages being Sent/received from/by your Computer
- Store and display the contents of the various protocol fields in the messages
- Passive program
  - never sends packets itself
  - no packets addressed to it
  - receives a copy of all packets (sent/received)

Packet Sniffer structure Diagnostic Tools.

- Tcpdump
  - E.g. tcpdump - enx host 10.129.41.2 -w exe3.out
- Wireshark
  - wireshark -r exe3.out

Description :

Wireshark.

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect individual packets. you can use Wireshark to inspect a suspicious program's network traffic, analyze the traffic flow on your network, or troubleshoot network problems.

Capturing and analysing packets using wireshark Tool

To filter, Capture, view, packets in Wireshark Tool

Capture 100 packets from the Ethernet : IEEE 802.3 LAN Interface and Save it.

Procedure

- Select Local Area Connection in Wireshark.
- Go to Capture option
- Select stop Capture automatically after 100 packe
- Then Click Start Capture.
- Save the packets.

1. Create a Filter to display only TCP/UDP packets, inspect the packets and provide the flow graph.

Procedure

- Select Local Area Connection in Wireshark.
- Go to Capture option
- Select Stop Capture automatically after 100 packet
- Then Click Start Capture.
- Search TCP packets in search bar.
- To see flow graph click Statistics Flow graph.
- Save the packets.

2. Create a Filter to display only ARP packets and inspect the packets. Procedure.

- Go to Capture option.
- Select stop Capture automatically after 100 packets
- Then click Start Capture.
- Search ARP packets in Search bar,
- Save the packets.

3. Create a Filter to display only DNS packets and provide the flow graph. Procedure.

- Go to capture option
- Select stop Capture automatically after 100 packets.
- Then click Start Capture.
- Search DNS packets in search bar.
- To see flow graph Click statistics Flow graph.
- Save the packets.

4. Create a Fitter to display only HTTP packets and inspect the packets Procedure.

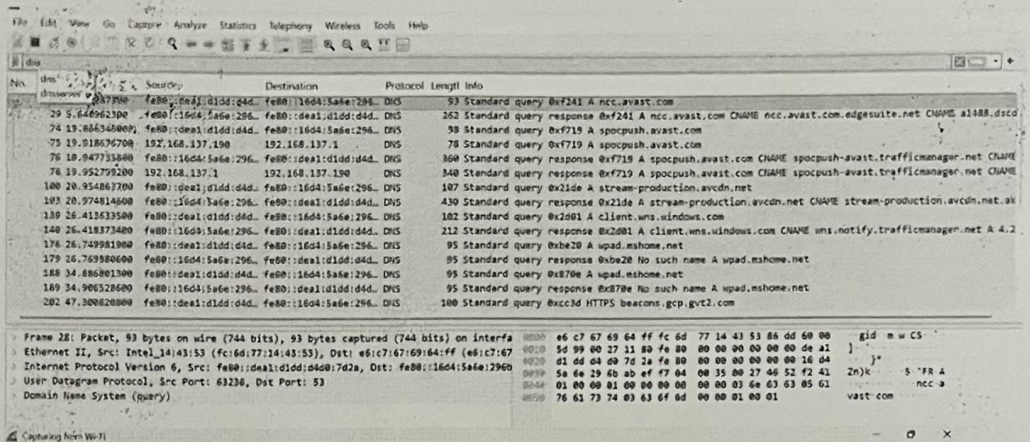
- Select Local Area Connection in Wireshark.
- Go to capture option
- Select stop capture automatically after 100 packets.
- Then click start Capture.
- Search HTTP packets in search bar,
- Save the packets.

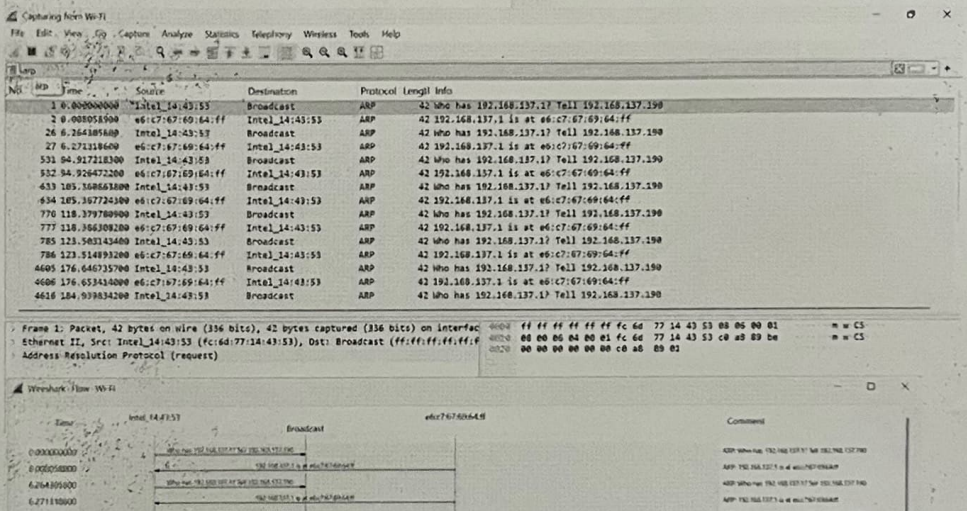
5. Create a Filter to display only IP/ICMP packets and inspect the packets. Procedure.

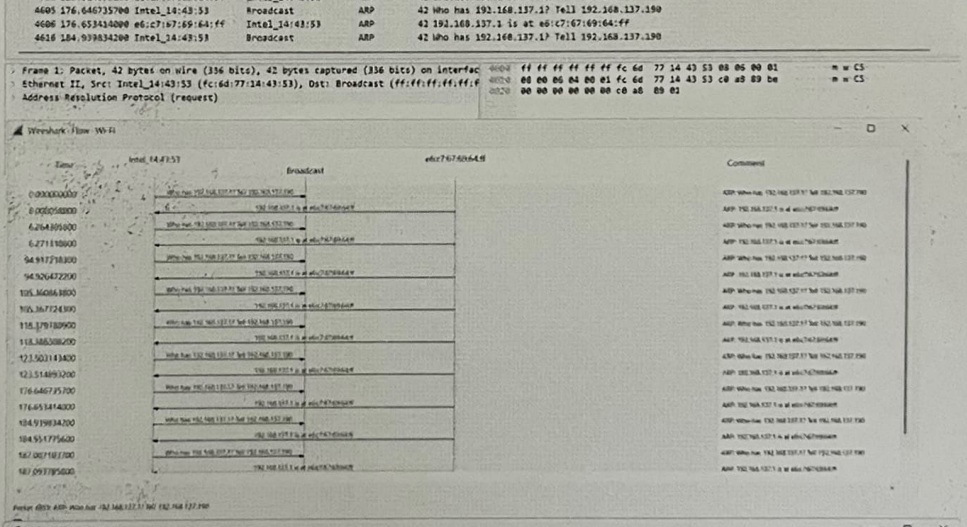
- Select Local Area Connection in Wireshark.
- Go to Capture option.
- Select stop Capture automatically after 100 packets
- Then click Start Capture.
- Search ICMP/IP packets in search bar.
- Save the packets.

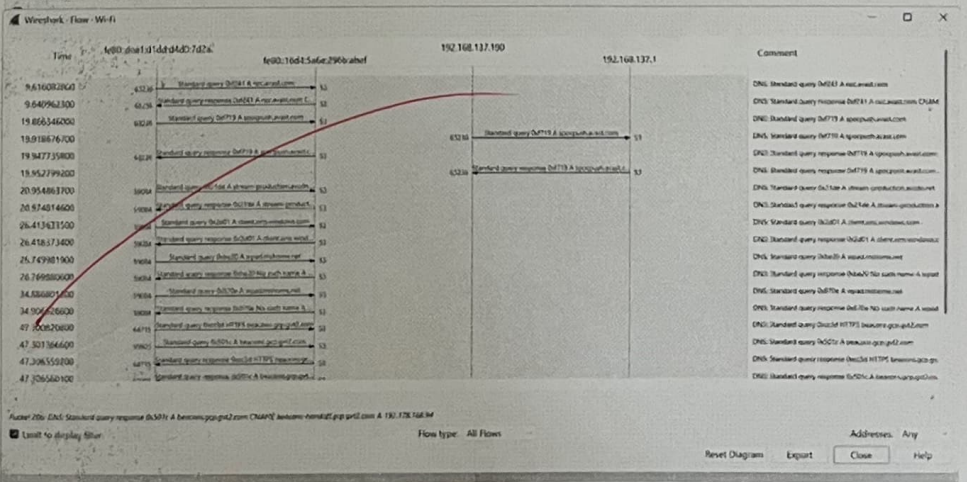
6. Create a Filter to display only DHCP packets and inspect the packets. Procedure.

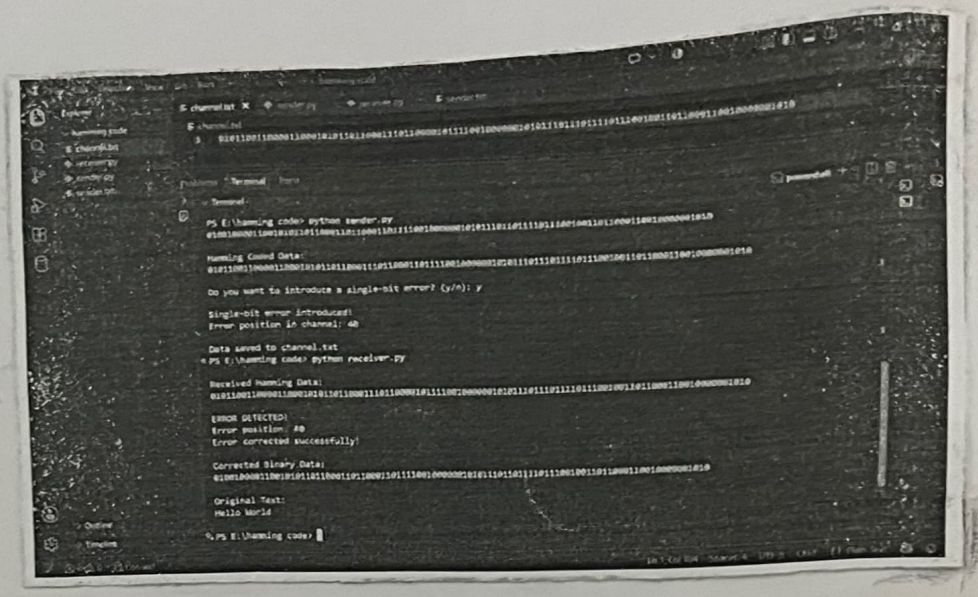
- Select the Local Area Connection in Wireshark.
- Go to capture option.
- Select Stop Capture automatically after 100 packets
- Then click Start Capture.
- Search DHCP packets in search bar.
- Save the packets.


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
1 0.000000000 fe80::de41:d1dd:d4d... fe80::16d4:5a6e:296... DNS 93 Standard query 0xf241 A ncc.avast.com
29 5.648962300 fe80::16d4:5a6e:296... fe80::dea1:d1dd:d4d... DNS 262 Standard query response 0xf241 A ncc.avast.com CNAME ncc.avast.com.edgesuite.net CNAME a1488.dscd
74 19.886546000 fe80::dea1:d1dd:d4d... fe80::16d4:5a6e:296... DNS 98 Standard query 0xf719 A spocpush.avast.com
75 19.918676700 192.168.137.190 192.168.137.1 DNS 78 Standard query 0xf719 A spocpush.avast.com
76 19.947735800 fe80::16d4:5a6e:296... fe80::dea1:d1dd:d4d... DNS 360 Standard query response 0xf719 A spocpush.avast.com CNAME spocpush-avast.trafficmanager.net CNAME
78 19.952799200 192.168.137.1 192.168.137.190 DNS 340 Standard query response 0xf719 A spocpush.avast.com CNAME spocpush-avast.trafficmanager.net CNAME
100 20.954861700 fe80::dea1:d1dd:d4d... fe80::16d4:5a6e:296... DNS 107 Standard query 0x21de A stream-production.avcdn.net
103 20.974814600 fe80::16d4:5a6e:296... fe80::dea1:d1dd:d4d... DNS 430 Standard query response 0x21de A stream-production.avcdn.net CNAME stream-production.avcdn.net.ak
139 26.413633500 fe80::dea1:d1dd:d4d... fe80::16d4:5a6e:296... DNS 102 Standard query 0x2d01 A client.wns.windows.com
140 26.418373400 fe80::16d3:5a6e:296... fe80::dea1:d1dd:d4d... DNS 212 Standard query response 0x2d01 A client.wns.windows.com CNAME wns.notify.trafficmanager.net A 4.2
176 26.749981900 fe80::dea1:d1dd:d4d... fe80::16d4:5a6e:296... DNS 95 Standard query 0xbe20 A wpad.mshome.net
179 26.769580600 fe80::16d4:5a6e:296... fe80::dea1:d1dd:d4d... DNS 95 Standard query response 0xbe20 No such name A wpad.mshome.net
188 34.886801300 fe80::dea1:d1dd:d4d... fe80::16d4:5a6e:296... DNS 95 Standard query 0x870e A wpad.mshome.net
189 34.906528600 fe80::16d4:5a6e:296... fe80::dea1:d1dd:d4d... DNS 95 Standard query response 0x870e No such name A wpad.mshome.net
202 47.300620800 fe80::dea1:d1dd:d4d... fe80::16d4:5a6e:296... DNS 100 Standard query 0xccd3d HTTPS beacons.gcp.gvt2.com
Frame 28: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interfa
Ethernet II, Src: Intel_14:43:53 (fc:6d:77:14:43:53), Dst: e6:c7:67:69:64:ff (e6:c7:67
Internet Protocol Version 6, Src: fe80::dea1:d1dd:d4d0:7d2a, Dst: fe80::16d4:5a6e:2960
User Datagram Protocol, Src Port: 63236, Dst Port: 53
Domain Name System (query)
0000 e6 c7 67 69 64 ff fc 6d 77 14 43 53 86 dd 60 00 gid m w CS
0010 5d 99 00 27 11 80 fe 80 00 00 00 00 00 00 de a1]
0020 d1 dd d4 d0 7d 2a fe 80 00 00 00 00 00 00 16 d4)*
0030 5a 6e 29 6b ab ef f7 04 00 35 00 27 d6 52 f2 41 Zn)k 5 'R A
0040 01 00 00 01 00 00 00 00 00 00 03 6e 63 03 63 85 61 ncc a
0050 76 61 73 74 03 63 6f 6d 00 00 01 00 01 vast com


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Length Info
1 0.000000000 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
2 0.005058900 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
26 6.264305800 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
27 6.271318600 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
531 94.917218300 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
532 94.926472200 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
633 105.168661800 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
634 105.367724300 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
770 118.379780900 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
777 118.386308200 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
785 123.503143400 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
786 123.514893200 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
4605 176.646735700 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
4606 176.653414000 e6:c7:67:69:64:ff Intel_14:43:53 ARP 42 192.168.137.1 is at e6:c7:67:69:64:ff
4616 184.939834200 Intel_14:43:53 Broadcast ARP 42 Who has 192.168.137.1? Tell 192.168.137.190
Frame 1: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interfac
Ethernet II, Src: Intel_14:43:53 (fc:6d:77:14:43:53), Dst: Broadcast (ff:ff:ff:ff:ff:f
Address Resolution Protocol (request)
0000 ff ff ff ff ff ff fc 6d 77 14 43 53 08 06 00 01 m w CS
0010 08 00 06 04 00 01 fc 6d 77 14 43 53 c0 a8 89 be m w CS
0020 00 00 00 00 00 00 c0 a8 89 01


Wireshark · Flow · Wi-Fi
Time
Intel_14:43:53
e6:c7:67:69:64:ff
Broadcast
Comment
0.000000000
0.005058900
6.264305800
6.271318600
94.917218300
94.926472200
105.168661800
105.367724300
118.379780900
118.386308200
123.503143400
123.514893200
176.646735700
176.653414000
184.939834200
184.951775600
187.087181700
187.097185600


Wireshark · Flow · Wi-Fi
Time
fe80::dea1:d1dd:d4d0:7d2a
fe80::16d4:5a6e:296b:abef
192.168.137.190
192.168.137.1
Comment
9.616082800
9.640962300
19.886546000
19.918676700
19.947735800
19.952799200
20.954861700
20.974814600
26.413633500
26.418373400
26.749981900
26.769580600
34.886801300
34.906528600
47.300620800
47.303364600
47.306559200
47.306560100
Limit to display filter
Flow type: All Flows
Addresses: Any
Reset Diagram
Export
Close
Help


PS E:\hamming code> python sender.py
Hamming Coded Data:
Do you want to introduce a single-bit error? (y/n): y
Single-bit error introduced!
Error position in channel: 40
Data saved to channel.txt
PS E:\hamming code> python receiver.py
Received Hamming Data:
[ERROR DETECTED]
Error position: 40
Error corrected successfully!
Corrected Binary Data:
Original Text:
Hello World
PS E:\hamming code>

Date: 24/8/26 Practical - 6

AIM : Write a program to implement error detection and correction using HAMMING Code Concept. Make a test run to input data stream and verify error Correction feature.

Error correction at Data Link Layer :

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the receiver. It is a technique developed by R.W. Hamming for error correction.

Create Sender program with below features.

1. Input to sender file should be a text of any length. Program should Convert the text to binary.
2. Apply hamming Code Concept on the binary data and add redundant bits to it.
3. Save this output in a file called channel.

Create a receiver program with below features.

1. Receiver program should read the input from channel file.
2. Apply hamming Code on the binary data to check for errors.
3. If there is an error, display the position of the error.
4. Else remove the redundant bits and Convert the binary data to ascii and display the output.

Sender Program.

```
#include <stdio.h>
#include <string.h>

int main() {
FILE *f = fopen("input.txt", "r");
FILE *c = fopen("channel.txt", "w");
char s[100], data[1000], code[1100];
int m=0, r=0, n, i, j, p, parity;
fgets(s, 100, f);
fclose(f);

for(i=0; s[i]; i++)
    for(j=7; j>=0; j--)
    data[m++] = ((s[i]>>j)&1) + '0';
While((1<<r) < m+r+1) r++;
  n = m+r; j=0;
for(i=1; i<=n; i++)
   code[i] = 1&(i-1))? data[j++]; '0';
for(p=0; p<r; p++) {
   parity = 0;
for(i=1; i<=n; i++)
   if(i&(1<<p)) parity ^= code[i] - '0';
code[1<<p] = parity + '0';

}
```

```
for (i=1; i<=n; i++) fprintf(c, "%c", code[i]);
  fclose(c);
printf("input: %s", s);
printf("Hamming Code saved in channel.txt\n");
  return 0;
```

Receiver Program

```
#include <stdio.h>
#include <string.h>
int main() {
  FILE *f = fopen("channel.txt", "r");
  char c[1100], data[1100];
  int n,i,p, error = 0, j = 0, x;
  fscanf(f, "%s", c);
  fclose(f);
  n = strlen(c);
  for(p=0); (1<<p)<=n; p++) {
      x = 0;
  for(i=1; j<=n; j++)
     if i & (1<<p)) x ^= c[i-1]
     if(x) error += 1<<p;
  }
     if(error) {
      printf("error at position: %d\n", error);
    c[error-1] = c[error-1]=='0')? '1': '0';
     printf(error corrected.\n");
  } else printf("No error deceted.\n");
```

```
for i = 1 ; i <= n ; i ++)
 if (i & (i-1)) data [ j++ ] = c [ i - 1 ] ;
 Printf (" Output ; " ) ;
 for (i = 0 ; i < j ; j + = 8) {
     x = 0 ;
  for ( P = 0 ; P < 8 ; P ++) x = x * 2 + data [i + P ] = '0' ;
     Printf (" % c", x ) ;
 }
   Printf ("\n") ;
    return 0 ;
}
```

Student observation.

Input (input, txt);

HELLO

Sender Output

Input ; HELLO

Hamming Code Saved in channel. txt.

Receiver Output (no error);

No error detected;

output ; HELLO.

25/8/26

Practical-7.

AIM : Write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

Sender program

```
# include <stdio.h>
# include <string.h>
# include <unistd.h>

int main () {

  Char msg [100], ack [20];
  int w, start = 0, i, n;

Print f (" Enter window size;");
 Scanf ("%d", &w);
 getchar ();
 Print f (" Enter message;");
 fgets (msg, 100, stdin);

while (start <n) {
 FILE * f = fopen ("sender_Buffer", "w");
  Print f ("\n Sending frames.\n");
 for (i = start; i < start + w && i < n; i++) {
  f printf (f, " frame NO : % d DATA : % c\n", i, msg [i]);
Print f (" Frame NO: %.d DATA : %.c \n", i, msg [i]);
}  f close (f);
Print f (" waiting for ACK...\n");
 Sleep (2);
```

```
f = fopen("Receiver_Buffer", "r");
fscanf(f, "%s %d", ack, &i);
fclose(f);
if (strcmp(ack, "Ack") == 0) {
  Printf("Ack %d received\n", i);
   start = i + 1;
} else {
   Printf("NAck %d received, Resending...\n", i);
   start. i;
 }
}
   Printf("\n All frames sent successfully\n");
   return 0;
}
```

Receiver Program.

```
# include <stdio.h>
# include <unistd.h>
int main() {
 FILE * f;
 int frame, expected = 0;
 Char data;
while(1) {
 f = fopen("Sender_Buffer", "r");
  if (f == NULL) continue;
if (fscanf(f, "Frame NO: %d DATA: %c", & frame,
& data) == 2) {
```

```
fclose(f);
  break;
}
fclose(f);
Printf("Received Frame %d : %c\n", frame, data);
f = fopen("Receiver_Buffer", "w");
if (frame == expected) {
   Printf("Ack %d Sent\n", frame);
fPrintf(f, "Ack %d", frame);
   expected++;
} else {
      Printf("NAck %d Sent\n", expected);
   fprintf(f, "NACK %d, expected;
}
    fclose(f);
     Sleep(2);
     break;
  }
   return
         0;
 }
```

Student Observation.

Input :

Window Size : 3

Message : ABCDE

Sender output :

Sending frames :

Frame NO : 0 DATA A

Frame No : 1 DATA B

Frame No : 2 DATA C

Ack 2 received

Sending frames :

Frame NO : 3 DATA : D

Frame NO : 4 DATA : E

Ack 4 received

All frames Sent Successfully.

Receiver Output :

Received Frame 0 : A

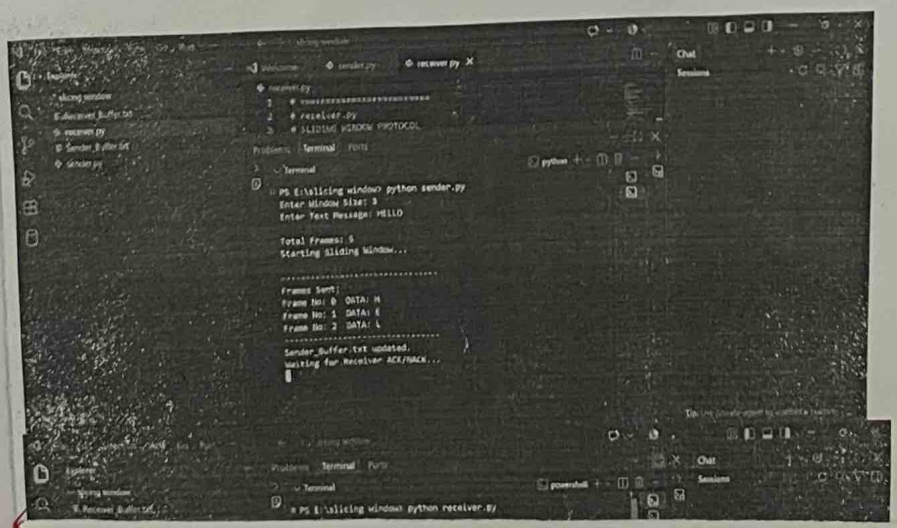
Ack 0 Sent

Received Frame 1 : B

Ack 1 Sent

Received Frame 2 : C

Ack 2 Sent.



18/2/26 PRACTICAL 8.

AIM :

To discover live hosts on a network using Nmap host-discovery scans - ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

Introduction.

Nmap ("Network Mapper") is a free, open-source network-scanning tool created by Gordon Lyon (Fyodor) and first released in 1997. It is the industry-standard tool for mapping networks, discovering live hosts, and indentifying the services running on them. Before Nmap can usefully scan ports on a target, it first needs to know whether the target is even online. Port-scanning an offline address wastes time and creates unnecessary network noise — so host discovery is always the first practical step of a network assessment.

This lab fouses on the first of two questions asked during reconnaissance :

1. which systems are up ? — answered in this room (host discovery).
2. what services are running on those systems ? — answered in the follow-on Nmap port-scanning rooms.

Nmap can determine whether a host is alive using four different protocols, one at each relevant layer of the TCP/IP stack:

- ARP (Link layer) – for hosts on the same local subnet
- ICMP (Network layer) – echo, timestamp, and address-mask requests.
- TCP (Transport layer) – SYN and ACK ping probes.
- UDP (Transport layer) – relies on an ICMP "Port unreachable" reply.

Lab setup on Try Hack ME

Follow these steps to prepare the lab environment before running any scan:

1. Open the room: https://tryhackme.com/room/nmap01 and log into your Try Hack Me account.
2. Read Task 1 (Introduction) and Task 2 (Subnetworks); use the embedded network simulator to answer the conceptual questions about ARP broadcast domains.
3. In Task 4 ("Enumerating Targets"), click start to deploy the target network – a small lab with one Windows host and one Ubuntu host. Allow up to two minutes for the machines to initialise.
4. Click start Attack Box has booted, open a Terminal inside it. If instructed by the room, download and apply the provided .OVPN configuration file so that Attack Box can reach the private lab subnet.

5. Once the Attack Box has booted, open a Terminal inside it. If instructed by the room, download and apply the provided .ovpn Configuration file so the Attack Box can reach the private lab subnet.

6. Note the CONNECTION IP / target subnet shown on the room's network diagram — you will substitute this into every nmap command below.

you are now ready to run each host-discovery technique against the lab subnet.

ICMP Host Discovery

ICMP Echo (-PE)

ICMP Timestamp (-PP)

ICMP Address Mask (-PM)

NMap's Default Host-Discovery Behaviour.

When no discovery option is specified, NMAP chooses automatically;

1. Privileged user, local Ethernet target → ARP requests.
- Privileged user, target outside the local network → ICMP Echo, TCP Ack to port 80, TCP SYN to Port 443, and ICMP Timestamp, combined.
- unprivileged user, target outside the local network - TCP three-way handshake (SYN) to ports 80 and 443.

Add -sn to any scan to perform host discovery only without following up with a port scan. Add -Pn to skip host discovery entirely and treat every target as online (useful when discovery probes are blocked but you still want to port-scan.

Masscan - A Faster Alternative

Masscan performs a similar discovery / Port sweep but transmits packets far more aggressively, trading some accuracy for raw speed on large ranges. It is not installed on the AttakBox by default (apt install masscan).

Its syntax mirrors Nmap's port options:

```
$ masscan 10.200.6.0/24 -p 443
$ masscan 10.200.6.0/24 -p 80,443
$ masscan 10.200.6.0/24 -p 22-25
```

arp-scan is a similar dedicated tool that performs only the ARP-based portion of host discovery, useful for a quick, low-noise sweep of the local subnet.


Nmap Live Host Discovery

7/9/26 Practical - 9.

AIM :- Implementation of SUBNETTING in CISCO PACKET TRACER Simulator.

Classless IP subnetting is a technique that allows for more efficient use of IP addresses by allowing for subnet masks that are not just the default masks for each IP class. This means that we can divide out IP address space into smaller subnets, which can be useful when we have a limited number of IP addresses but need to create multiple networks.

CREATING A NETWORK TOPOLOGY :

The first step in implementing classless IP subnetting is to create a network topology in Packet Tracer. To create a network topology in Packnet Tracer, select the "New" button in the top left corner, then select "Network" and "Generic". This will create a blank network topology that we can use to add devices.

ADDING THE DEVICES :

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches, and PCs. To add a device, select the device from the bottom left corner and drag it onto the network topology. Then connect the devices by dragging a cable from one device's port to another device's port.

SUBNETTING :

To subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 address for end devices, the switch, and router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host addresses each.

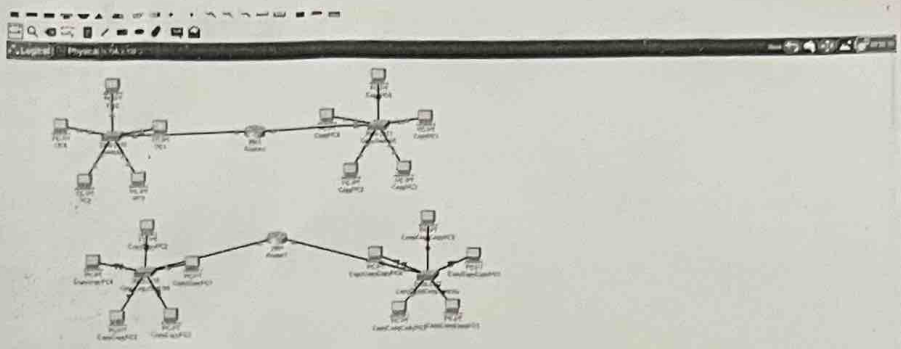
CONFIGURING THE DEVICES :

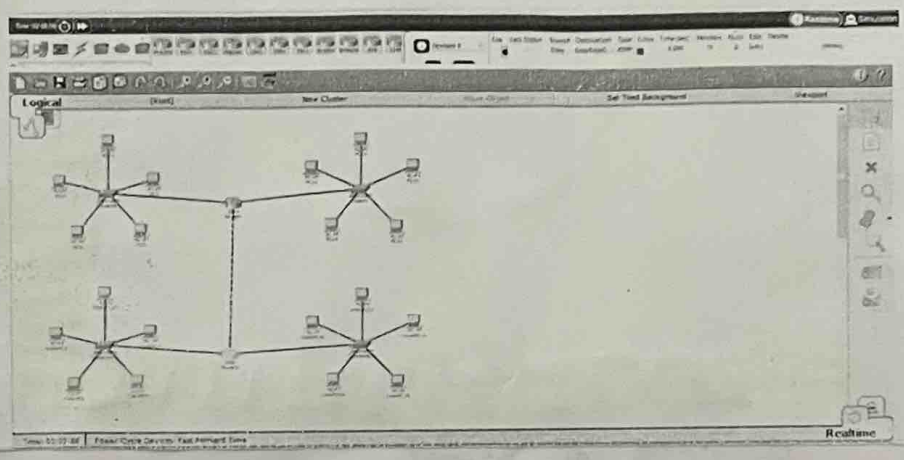
Now that we have added our devices and connected them, we can start configuring them. we will start by configuring the router. Right-click on the router and select "CLI". This will open the command-line interface (CLI) for the router. In the CLI, enter the following commands.

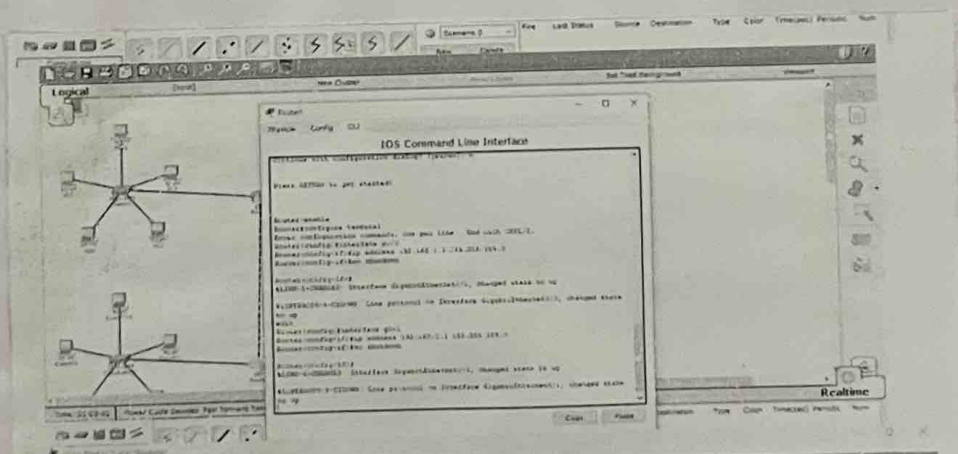
```
# enable
# configure terminal
# interface Fast Ethernet 0/0
# ip address [IP address] [subnet mask]
# no shutdown
# exit
interface Fast Ethernet 0/1
ip address (IP address) (subnet mask.
no shutdown
exit.
```

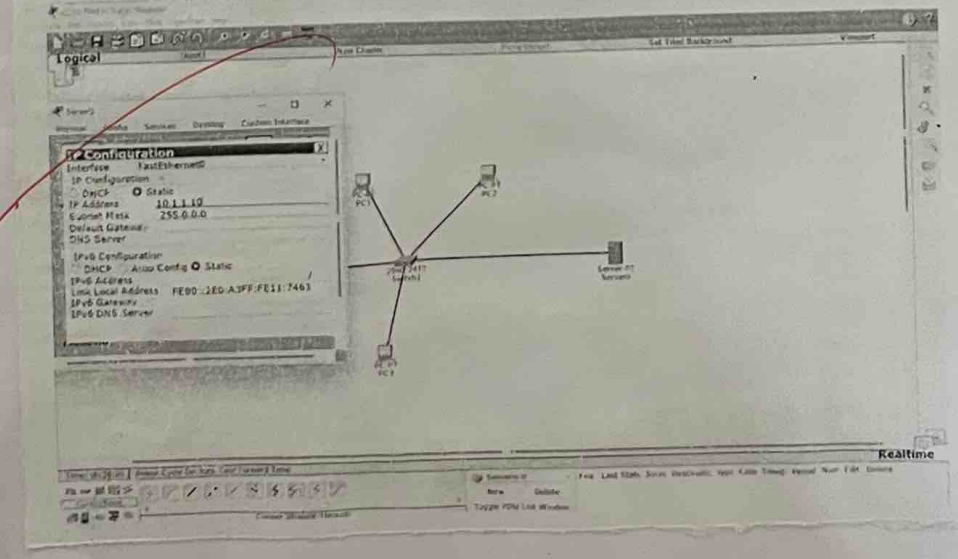
Testing The network :

Now that our network topology is configured, we can test the network. Open a command prompt on each PC and try to ping the other PC. If the ping is successful, then the network is functioning properly.




Logical
Realtime


Logical
IOS Command Line Interface
Realtime


Logical
Fa0 Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IP Address 10.1.1.10
Subnet Mask 255.0.0.0
Default Gateway
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address FE80::2E0:A3FF:FE11:7463
IPv6 Gateway
IPv6 DNS Server
Realtime

14/9/26

Practical - 10.

AIM : a) Internetworking with routers in CISCO PACKET TRACER Simulator.

Design and Configure a simple internetwork using a router.

In this network, a router and 2 PCs are used. Computers are connected with routers using a copper straight-through cable. After forming the network, to check network connectivity a simple PDU is transferred from PC0 to PC1.

Procedure :

Step-1. (Configuring Router1) :

1. Select the router and open CLI.
2. Press ENTER to start Configuring Router 1.
3. Type enable to activate to privileged mode.

Step - 2. (Configuring PCs) :

1. Assign IP addresses to every PC in the network.
2. Select the PC, go to the desktop and select IP Configuration and assign an IP address. Default gateway, Subnet Mask.
3. Assign the default gateway of PC0 as 192.168.10.1.
4. Assign the default gateway of PC1 as 192.168.20.1

Step - 3 (Connecting PCs with Router) :

1. Connect Fast Ethernet 0 port of PC0 with Fast Ethernet 0/0 port of Router 1 using a copper Straight-through cable.
2. Connect Fast Ethernet 0 port of PC1 with Fast Ethernet 0/1 port of Router 1 using a copper Straight-through cable.

Designed Network topology.

Simulation of Designed Network Topology.

Sending a PDU from PC0 to PC1:

Acknowledgment from PC1 to PC0:

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AIM:- b) Design and configure an internetwork using wireless router, DHCP Server and internet cloud.

Addressing Table.

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC	Ethernet 0	DHCP		192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	
Wireless Router	internet	DHCP		
Cisco.com in Server	Ethernet 0	208.67.220.220	255.255.255.0.	
Laptop	wireless 0	DHCP		

Objectives:

Part 1: Build a Simple Network in the Logical Topology workspace.

Part 2: Configure the network Devices.

Part 3: Test Connectivity between network Devices.

Part 4: Save the File and close Packet Tracer.

Part 1: Build a Simple Network in the Logical Topology workspace.

Step 1: Launch Packet Tracer.

Part 2: Configure the Network Devices.

Step 1: Configure the wireless router.